

**Operating Pilot Plant via Station****11/28/2022*****Rev: 1******ABSTRACT:***

The purpose of this document is to provide prospective MUPP users information on the plant, such that, a level of competence sufficient to operate the MUPP is obtained. Within this document the following topics are covered:

- A brief introduction to the purpose of Experion Station.
- An overview of the information and interactive elements on Station screens.
- A guide to navigating the multitude of MUPP screens.
- A summary of the MUPP screens and the information and interactive elements contained therein.
- The purpose and function of Station security levels and access.
- A comprehensive guide to the MUPP controllers.
- A list of MUPP interlocks.
- A guide to using Station trend pages and trend groups.

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Document Control and Revision History

Table 1: Document Control and Revision History

Revision	Date	Author(s)	Reviewed by	Comments/Details
0	2010	Earl Hopkinson		
1	2022	Nicholas O’Keeffe		Retains PID FB Algorithms, and Display Hierarchy and Control Levels sections in appendices A and B, respectively. All other sections have been remade or replaced.

Definitions

BM	Ball Mill
BMP	Ball Mill Pump
BMT	Ball Mill Tank
CFT	Cyclone Underflow Tank
CM	Control Module
CRP	Cyclone Recycle Pump
CSTR	Continuously Stirred Tank Reactor
CUFP	Cyclone Underflow Pump
DV	Disturbance Variable
EPKS	Experion Process Knowledge System
Err	Error
FCV	Flow Control Valve
FDP	Flow Disturbance Pump
FP	Feed Pump
FT	Flow Transmitter
Gff	Feedforward Controller
HMI	Human Machine Interface
INT	Interlock
Kc	Controller Gain
LM	Lamella Tank
LUP	Lamella Underflow Pump
MUPP	Murdoch University Pilot Plant
MV	Manipulated Variable
NLT	Non-Linear Tank
NT	Needle Tank
NTP	Needle Tank Pump
OOC	Out of Commission
OP	Operating Point
OPC	Open Platform Communication
PID	Proportional, Integral, Derivative
PP	Product Pump
PV	Process Variable
SP	Set Point
ST	Storage Tank
Td	Controller Derivative Time
Ti	Controller Integral Time
TT	Temperature Transmitter
VSD	Variable Speed Drive

1.0 Purpose

The purpose of this document is to introduce prospective users of the Murdoch University Pilot Plant (*MUPP*) to Experion Station. Experion Station is the Honeywell proprietary software environment tasked with presenting the MUPP to plant operators. With it, users can interact with and control the numerous elements that make up the MUPP. This document aims to outline all the knowledge necessary for a MUPP user to gain a level of competency sufficient to operate the plant.

2.0 Scope

As previously mentioned, this document covers all the information necessary for a MUPP user to gain a level of competency sufficient to operate the plant; this includes:

- A brief introduction to the purpose of Experion Station.
- An overview of the information and interactive elements on Station screens.
- A guide to navigating the multitude of MUPP screens.
- A summary of the MUPP screens and the information and interactive elements contained therein.
- The purpose and function of Station security levels and access.
- A comprehensive guide to the MUPP controllers.
- A list of MUPP interlocks.
- A guide to using Station trend pages and trend groups.

3.0 Quick Start Guide

3.1 About Station

Experion station is the software package responsible for presenting HMI screens designed in the Experion HMI development tool. As such, the Station is responsible for presenting controller variables and parameters from the control building environment in a format that a plant user can navigate and interact with efficiently and in a straightforward manner.

Experion station provides all the necessary tools and information to operate the MUPP. Further, Station can tune and configure the plant's many controllers and view or export plant trends, alarms, and other historical data.

3.2 Station Display Screen Information

When Experion Station is first opened, it will default to page 301, the MUPP home screen (Figure 1). From this page, all relevant pages can be accessed; it is worth exploring the three information screens accessible by clicking on the three buttons on the left of the home page (Figure 1). Figure 2 shows the most relevant on-screen toolbars for the navigation and use of the MUPP. Figure 3 shows the most pertinent interactive elements for the operation of the MUPP; **pay particular attention to the 'Operation' checkbox, which must be checked before plant operation can occur.**

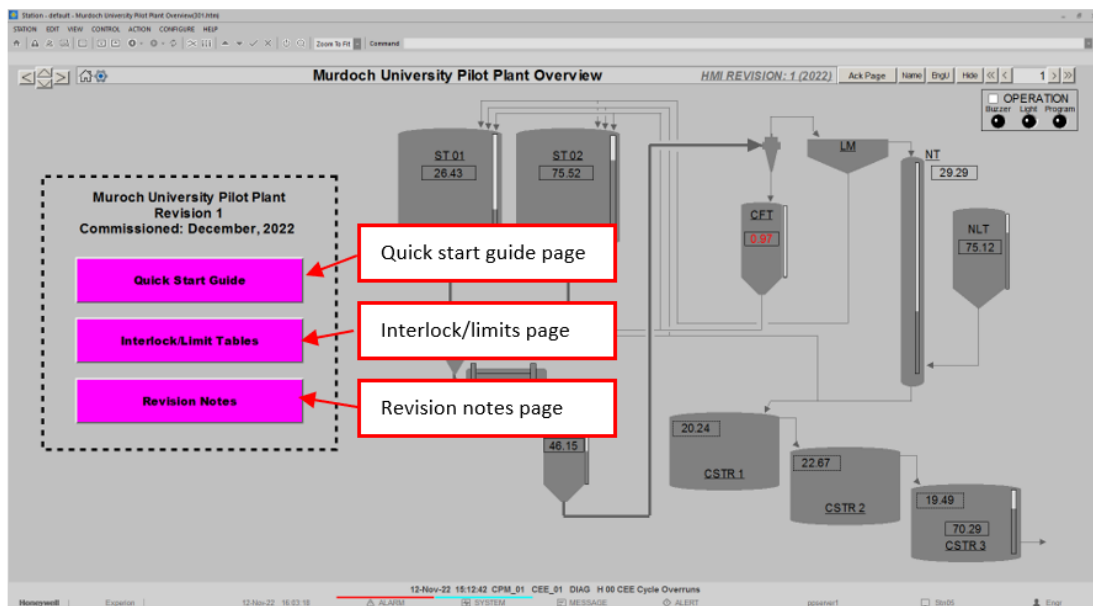


Figure 1: MUPP station screen 301, the MUPP home screen. The three pushbuttons on the left take the user to the quick start guide, the interlock/limits page, and the revision notes page, respectively.

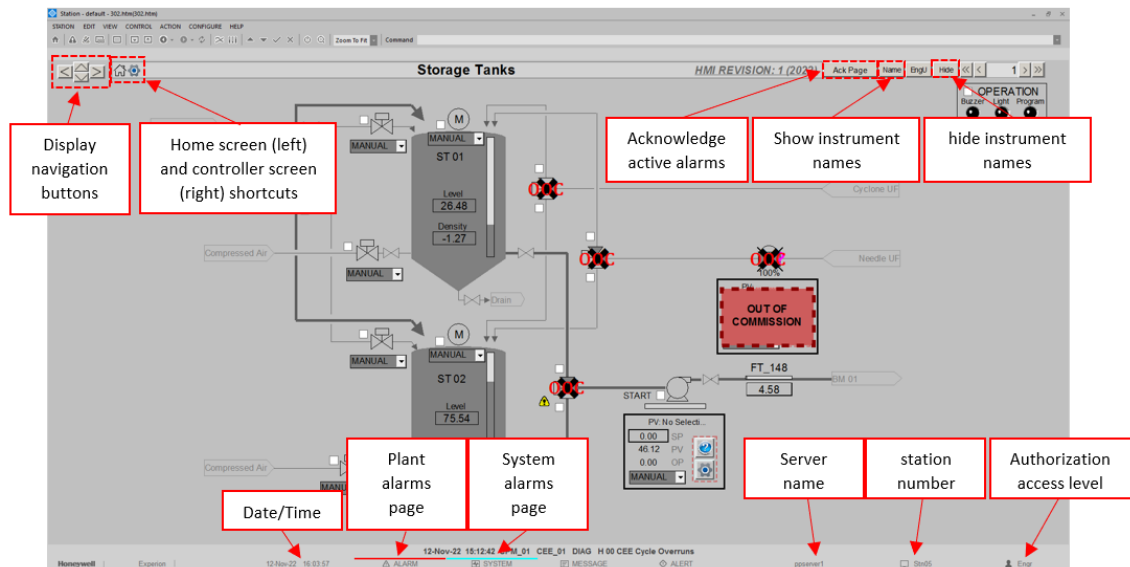


Figure 2: Relevant toolbar information is displayed on every screen in the MUPP HMI; highlighted information shows the most relevant in-screen toolbar elements for the operation and navigation of the MUPP.

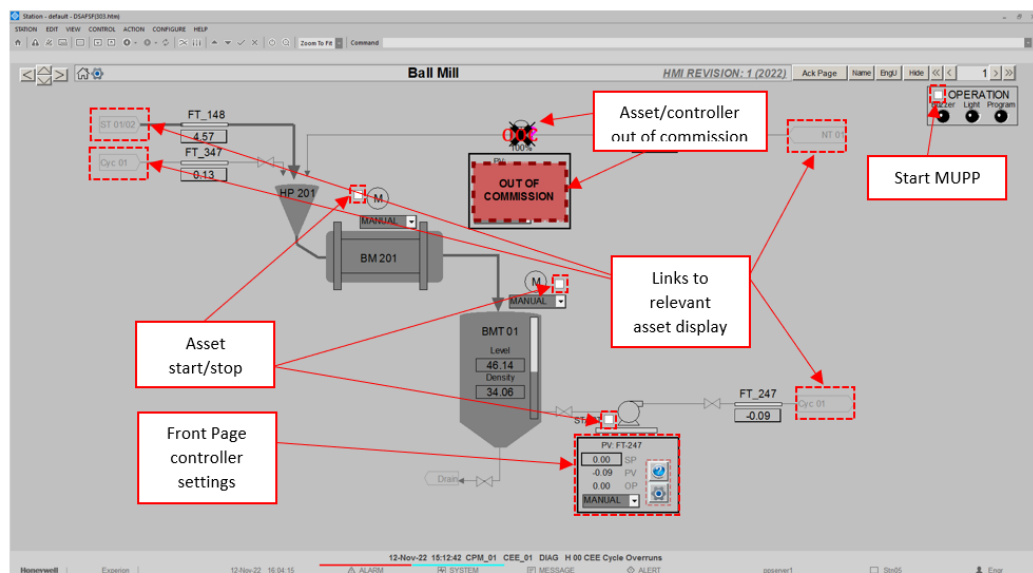


Figure 3: The critical elements of the MUPP HMI.

3.3 Navigating the MUPP HMI

The following sections outline navigation of the MUPP station screens and the information available on each.

3.3.1 The Home Screen

It is worth noting that all screens that require interaction for complete control of the MUPP can be accessed directly from the home screen (301); this can be done by clicking on the title of the asset on the home screen (*Figure 4*) where:

- **ST01** takes the user to the Storage Tanks Page (302)
- **ST02** takes the user to the Storage Tanks Page (302)
- **BM** takes the user to the Ball Mill Page (303)
- **BMT** takes the user to the Ball Mill Page (303)
- **CFT** takes the user to the Cyclone Page (304)
- **LM** takes the user to the Cyclone Page (304)
- **NT** takes the user to the Needle Tank Page (305)
- **NLT** takes the user to the Needle Tank Page (305)
- **CSTR 1** takes the user to the CSTR Bank Page (306)
- **CSTR 2** takes the user to the CSTR Bank Page (306)
- **CSTR 3** takes the user to the CSTR Bank Page (306)

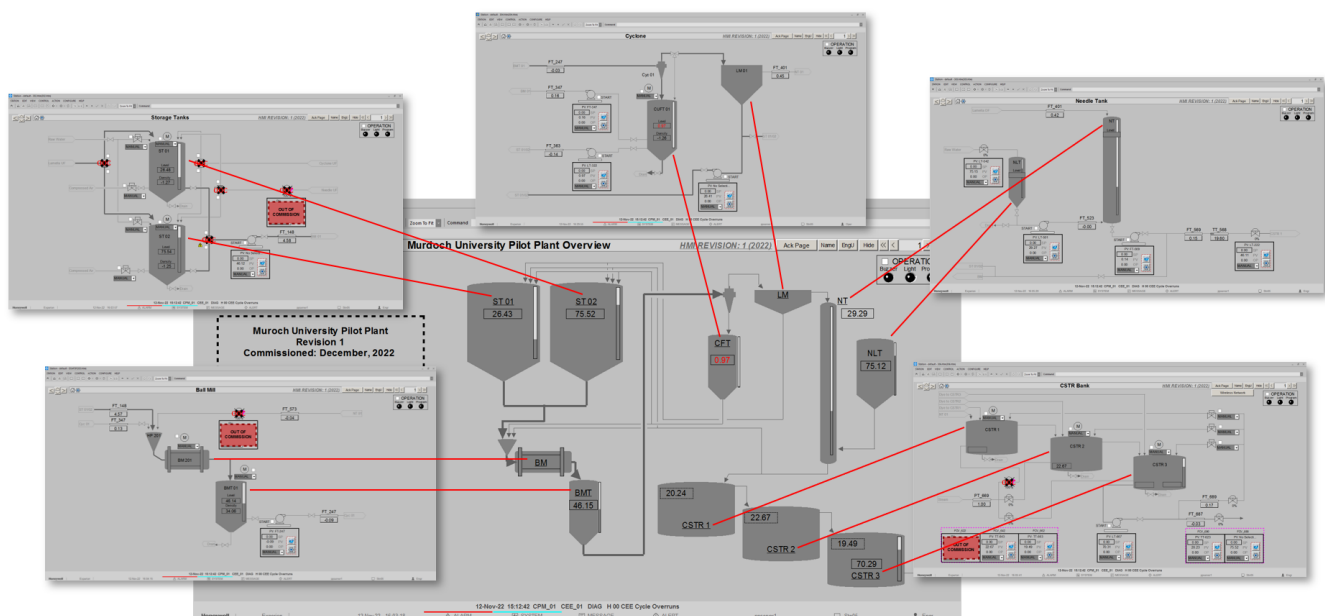


Figure 4: Each page relevant to the operation of the MUPP can be accessed directly from the home page (301) by selecting the asset's title.

3.3.2 Pages

301 – MUPP Main Page (Revision 1, 2022)

Page 301 (Figure 5) gives an overview of the MUPP, excluding the Dye Tank and wireless network pages, which are currently OOC. As mentioned previously, three user pages can be accessed via the magenta pushbuttons on the left-hand side of the screen, and each of the assets shown on the home page can be linked to by clicking on the asset title. The home screen can be returned from any page by selecting the home shortcut by the screen title (Figure 2).

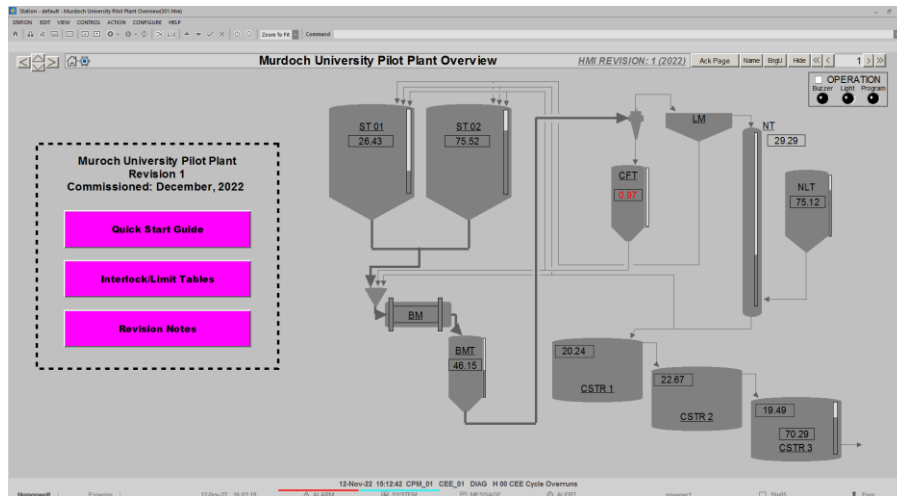


Figure 5: MUPP station display 301 – The home screen.

302 – Storage Tanks (Revision 1, 2022)

Page 302 (Figure 6) shows the storage tanks and feed pump (FP-141); several valves are present; however, all but the raw water and compressed air feeds into the storage tanks are OOC. From the storage tank page, FT-148 can be viewed, and the FP-141 controller can be accessed.

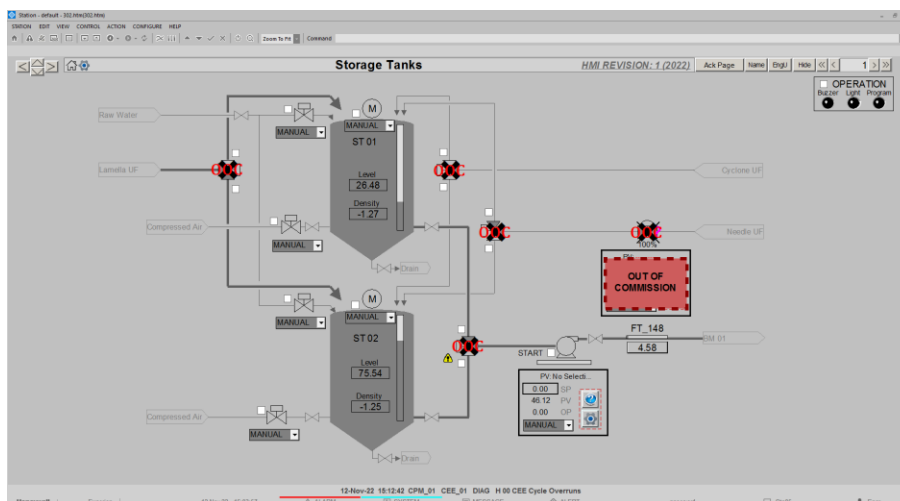


Figure 6: MUPP station display 302 – The storage tank screen.

303 – Ball Mill Page (Revision 1, 2022)

Page 303 (Figure 7) shows the ball mill, ball mill tank and ball mill tank underflow pump (BMP-241), there is one valve present, but this is OOC. From the ball mill screen, FT-247 and FT-573 can be viewed. From this page, the BMT-241 controller popup can be accessed, and its setpoint and mode of operation configuration can be set.

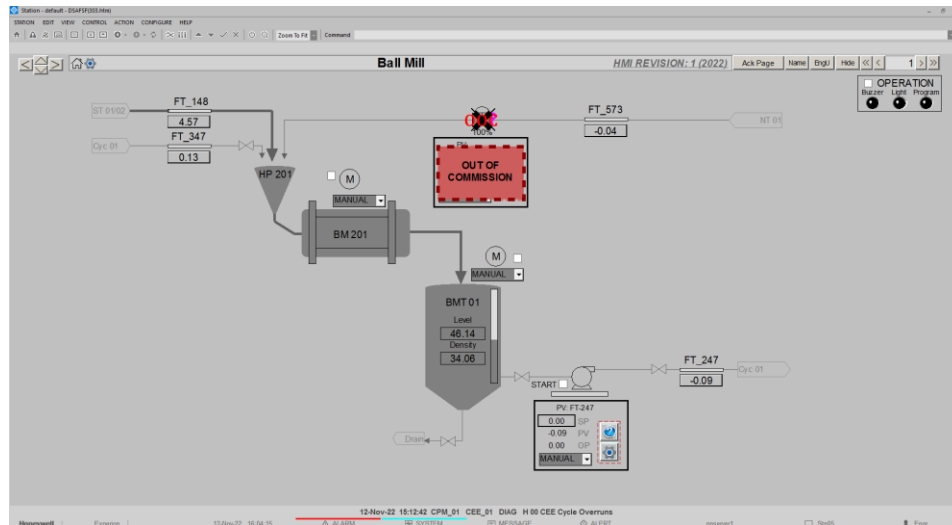


Figure 7: MUPP station display 303 – The ball mill screen.

304 – Cyclone Page (Revision 1, 2022)

Page 304 shows the cyclone underflow tank, lamella tank, cyclone recycle pump (CRP-341), cyclone underflow pump (CUFP-361), and lamella underflow pump (LUP-421). The outputs of FT's 247, 347, 363, and 401 can be read from the cyclone page. The controllers associated with CRP-341, CUFP-361, and LUP-421 can be accessed, and the configuration of their setpoint and operation can be set.

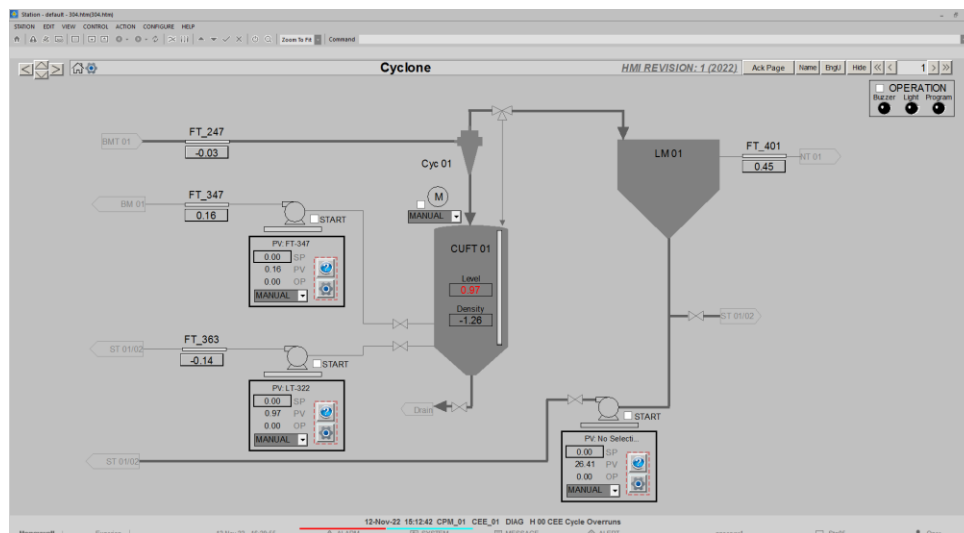


Figure 8: MUPP station display 304 – The cyclone screen.

305 – Needle Tank Page (Revision 1, 2022)

Page 305 (Figure 9) shows the nonlinear tank, needle tank, feed disturbance pump (FDP-521), needle tank underflow pump (NTP-561), and FCVs 541 and 570. From the needle tank page, the outputs of FT's 401, 523, 569, and TT-568 can be read. Further, the controllers associated with FDP-521, NTP-561, and FCV's 541 and 570 can be accessed, and their setpoints and mode of operation can be configured.

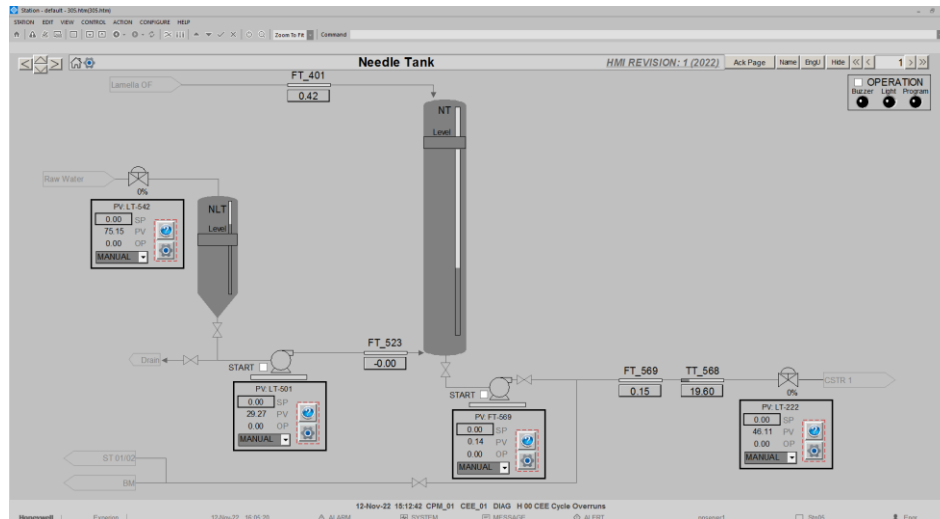


Figure 9: MUPP station display 305 – The needle tank screen.

306 – CSTR Bank (Revision 1, 2022)

Page 306 (Figure 10) shows CSTRs 1, 2, and 3, the product pump (PP-681), FCV's 622 (OOC), 642, 662, 688, and 690. From the CSTR bank page, the outputs of TTs 623, 643, 663, FTs 687, 689, and LT-667 can be read. Further, the controllers associated with PP-681 and FCVs 642, 662, 688, and 690 can be accessed, and their setpoints and mode of operation can be configured.

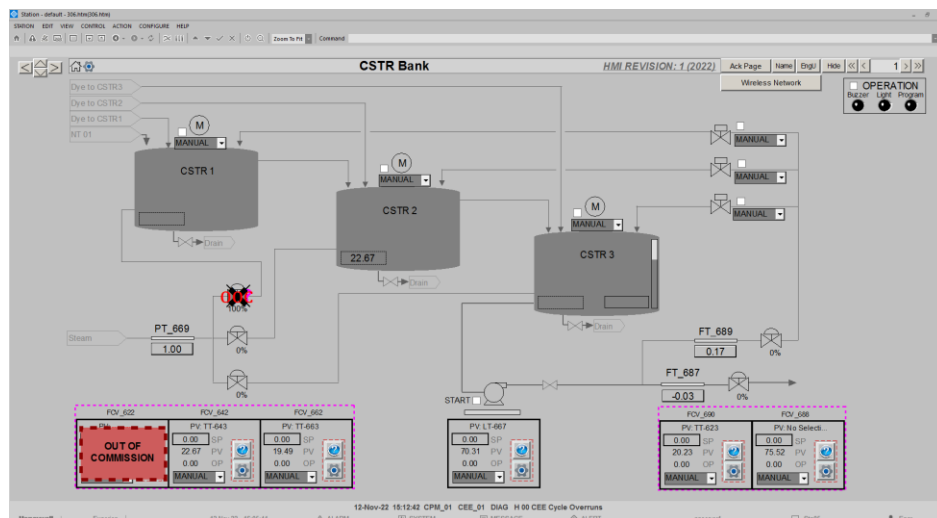


Figure 10: MUPP station display 306 – The CSTR bank screen.

307 – Dye System (Revision 0, ~2012)

Page 307 (Figure 11) shows the dye tank and dye pump, the function of these assets is unknown as they have not been used in some time and were not interacted with in the 2022 MUPP revision. Further investigation of these assets is recommended before use is attempted.

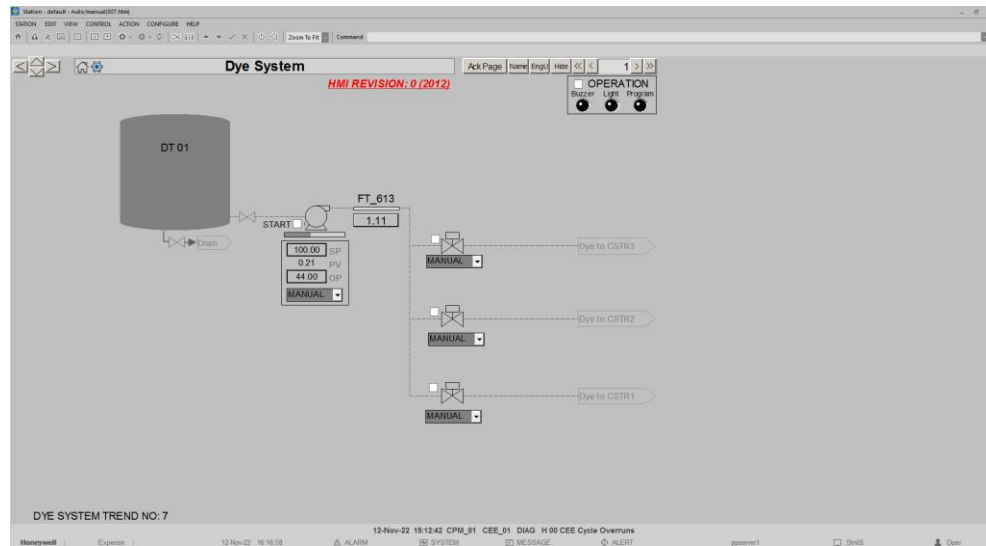


Figure 11: MUPP station display 307 – The dye system screen (From revision 0 programming ~2012).

308 – Drive Operating Conditions (Revision 0, ~2012)

Page 308 (Figure 12) shows the operating conditions of the nine variable speed drives tasked with operating the pumps in the MUPP; this page is a read-only page with no interactive elements.



Figure 12: MUPP station display 308 – The drive operating conditions screen (From revision 0 programming ~2012).

401 – Automatic Defined Settings (Revision 0, ~2012)

Page 401 (Figure 13) shows the 'Automatic Defined Settings' of the MUPP; these elements' functions are unknown as they have not been used in some time and were not interacted with in the 2022 MUPP revision. Further investigation of these elements is recommended before use is attempted.

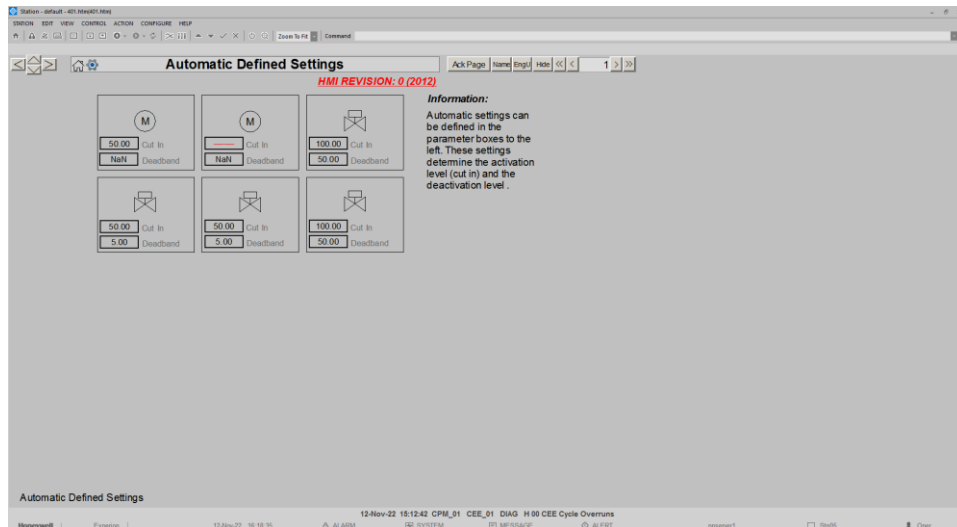


Figure 13: MUPP station display 401 – The 'Automatic Defined Settings' screen (From revision 0 programming ~2012).

402 – MUPP Control Loop Summary (Revision 1, 2022)

Page 402 (Figure 14) shows the front panel controller configuration tools for each MUPP controller; further, the full controller configuration popup windows can be accessed for each controller displayed. It is worth noting that this page can be accessed from any of the MUPP screens via the controller screen shortcut by the title (Figure 2).

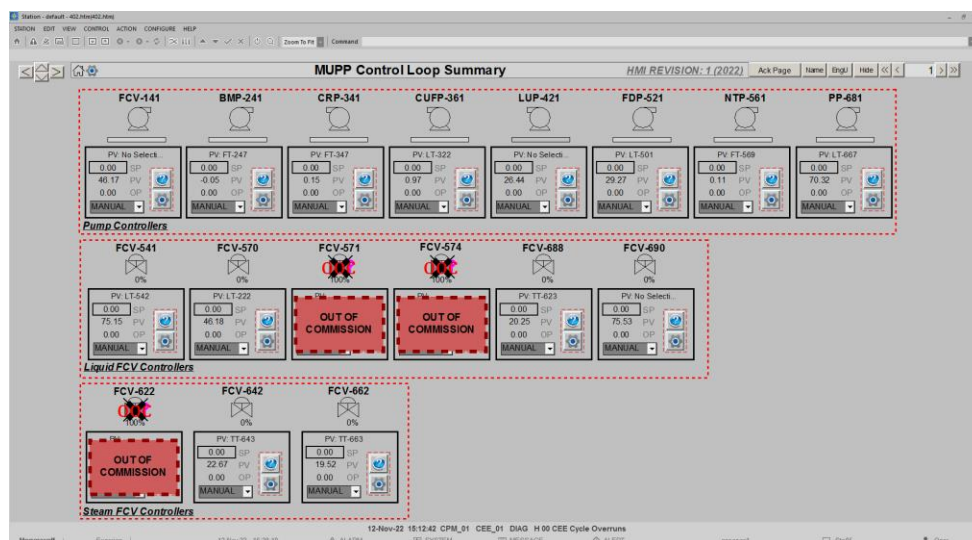


Figure 14: MUPP station display 402 – The MUPP control loop summary screen.

403 – Program Settings (Revision 0, ~2012)

Page 403 (Figure 15) shows the program settings relating to the maintenance and troubleshooting programs added in 2012. The function of these elements is unknown as they have not been used in some time and were not interacted with in the 2022 MUPP revision. Further investigation of these elements is recommended before use is attempted.

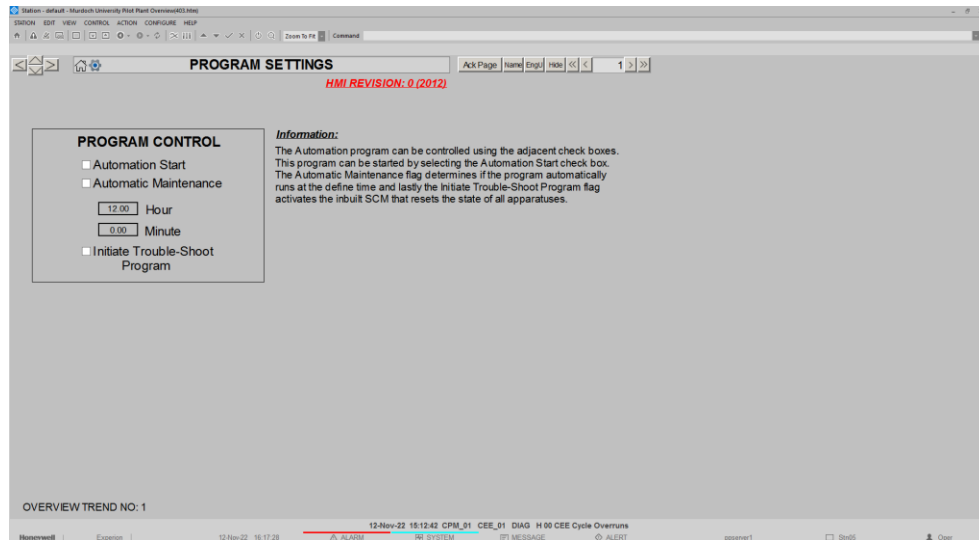


Figure 15: MUPP station display 403 – The program settings screen (From revision 0 programming ~2012).

404 – CSTR Bank – Wireless Network (Revision 0, ~2012)

Page 404 (Figure 16) shows the elements associated with the wireless network surrounding the CSTR bank. The function of these elements is unknown as they have not been used in some time and were not interacted with in the 2022 MUPP revision. Further investigation of these elements is recommended before use is attempted.

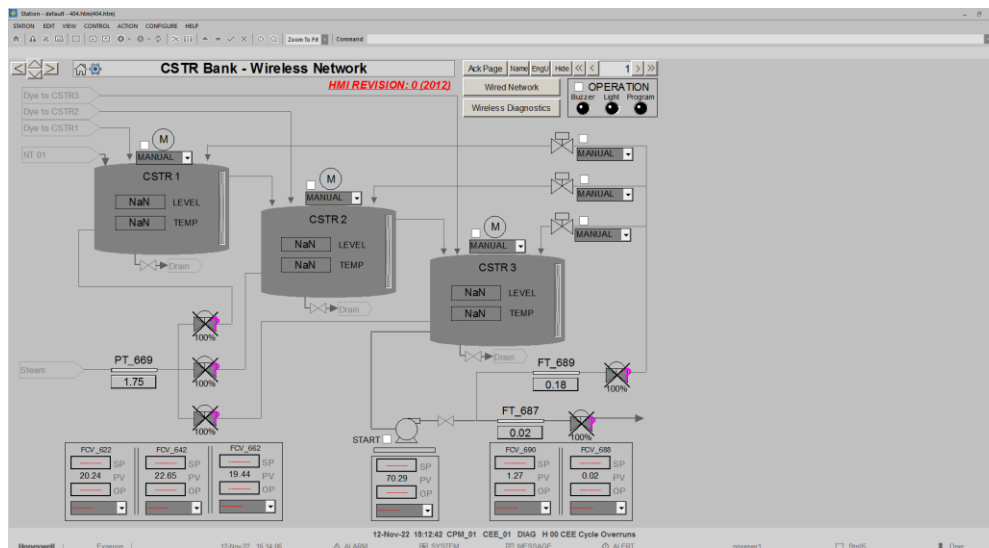


Figure 16: MUPP station display 404 – The CSTR bank, wireless network screen (From revision 0 programming ~2012).

405 – Wireless Network Diagnostics (Revision 0, ~2012)

Page 405 (Figure 17) shows the hardware elements associated with the wireless network. The function of these elements is unknown as they have not been used in some time and were not interacted with in the 2022 MUPP revision. Further investigation of these elements is recommended before use is attempted.

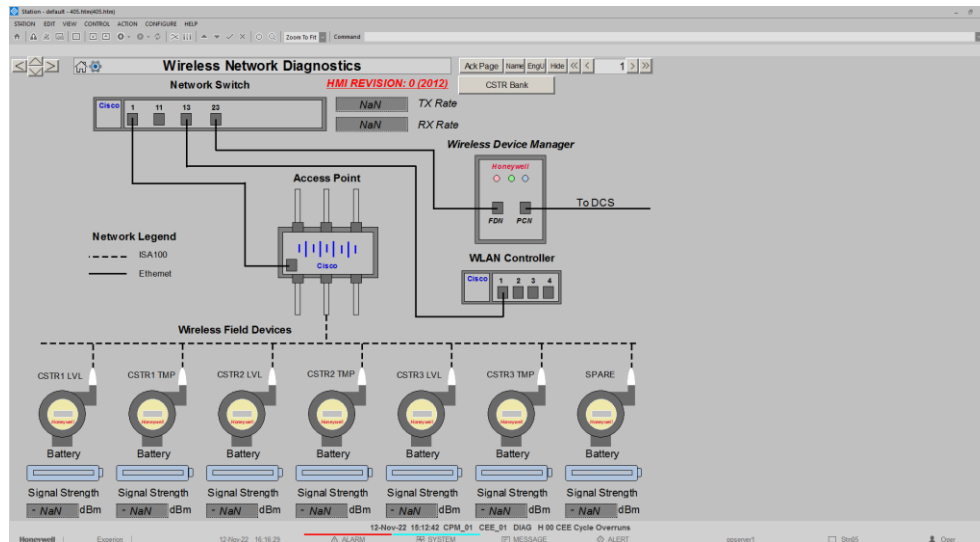


Figure 17: MUPP station display 405 – The wireless network diagnostics screen (From revision 0 programming ~2012).

3.4 Security Levels

Program security levels indicate the clearance of the MUPP user and correspond to the level of HMI and MUPP configuration permitted by a particular user. The security levels are as follows:

1. Operator
2. Supervisor
3. Engineer
4. Manager

Where the higher the level, the more access the user has. Several system settings, most notably in the controller configuration screens, are locked to users under the engineer security level. Select the authorization access level (Figure 2) and enter the password corresponding with the appropriate access level to gain access to a higher access level. Passwords are made available by the technical staff in charge of the MUPP.

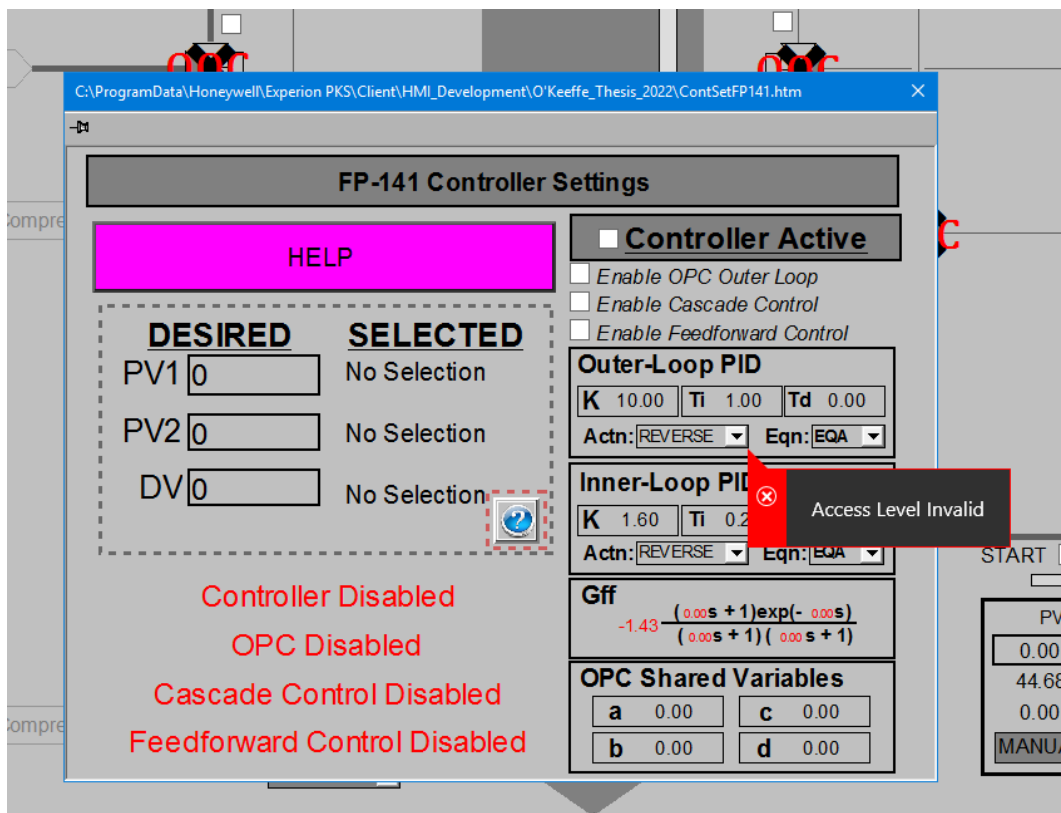


Figure 18: An instance of configuration being blocked due to an insufficient access level.

3.5 Using MUPP Controllers

The MUPP controllers were designed modularly; that is, the format of each of the controllers is identical, save for the title bar. The modularity of the implemented controllers was an intentional design feature to simplify the navigation and configuration of each of the fourteen implemented system controllers (*Figure 19*). The following subsections outline the structure and function of the front-page controller blocks and controller popup menus.

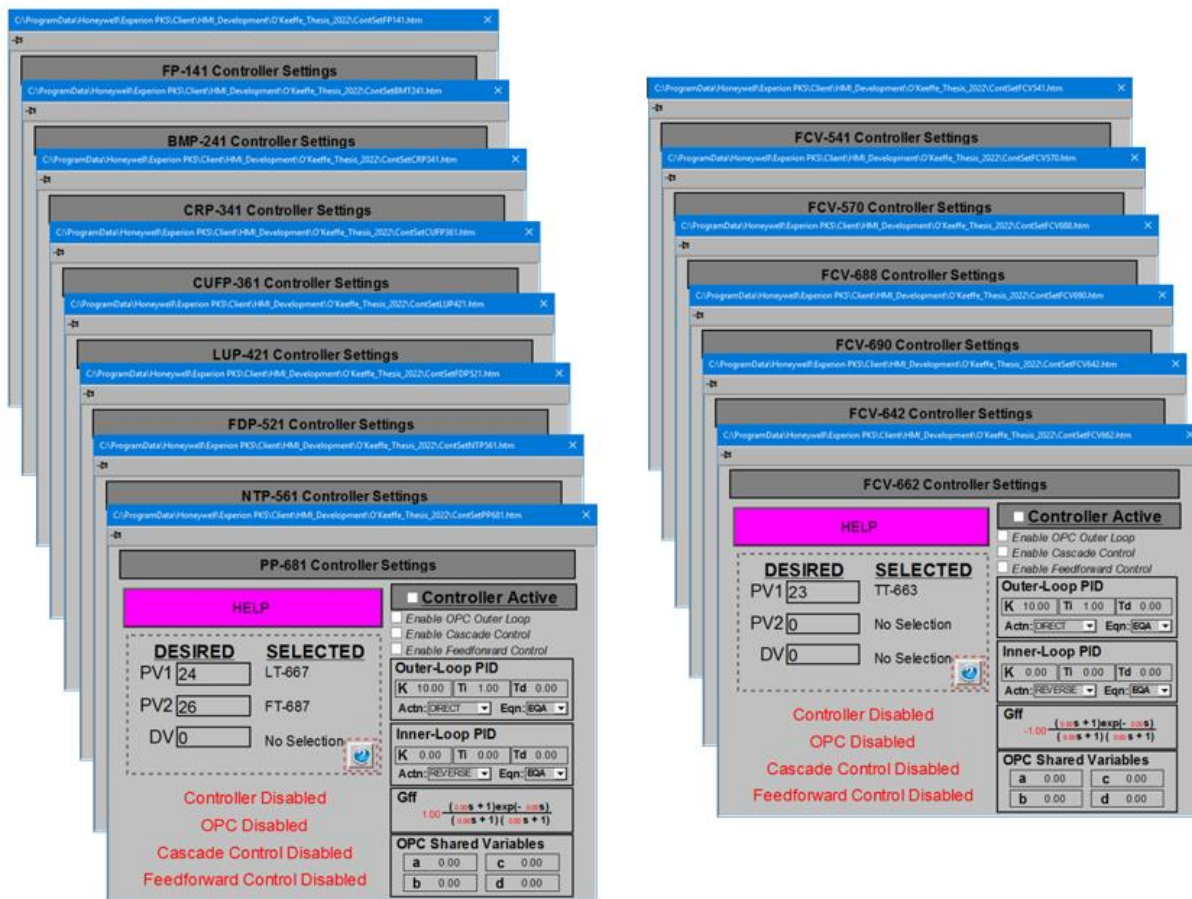


Figure 19: The MUPP system controllers.

3.5.1 Front-Page Controller Block

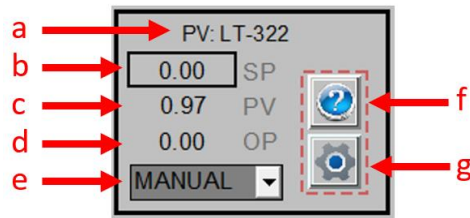


Figure 20: An instance of a front-page controller block.

The following descriptions refer to Figure 20:

- a. **PV selected display** – This shows the current outer-loop PV set, i.e., the controlled parameter.
NOTE: This is a display only and is configured from the controller popup menu.
- b. **Setpoint entry field** – In automatic mode (set by e.), this is a configurable field and sets controller setpoint; in manual mode (set by e.), this is a configurable field and sets the controller manipulated variable.
- c. **PV display** – This shows the value of the current outer loop PV.
- d. **OP/MV display** – This shows the value of the controller MV.
- e. **Operation mode selection** – This field is used to configure the controller's operating mode; the possible values are MANUAL and AUTOMATIC modes.
- f. **Front-Page controller help** – This pushbutton pulls up the front-page help popup screen.
- g. **Controller popup menu** – This pushbutton pulls up the controller popup menu.

3.5.2 Controller Popup Menu

Figure 21 shows an example of a controller settings popup menu, in this case, for FP-141. This section will outline the different fields and parameters configurable from the controller settings popup menu.

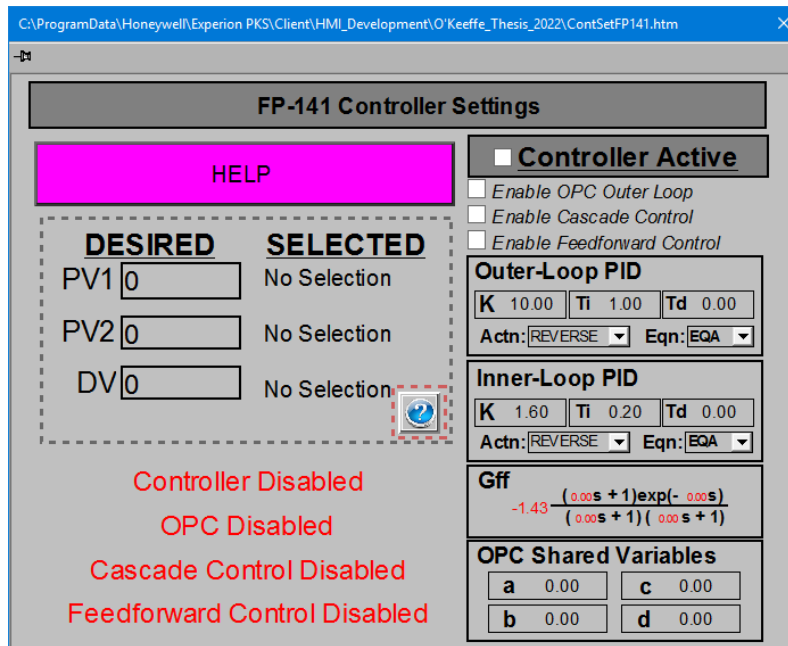


Figure 21: FP-141 controller settings popup menu.

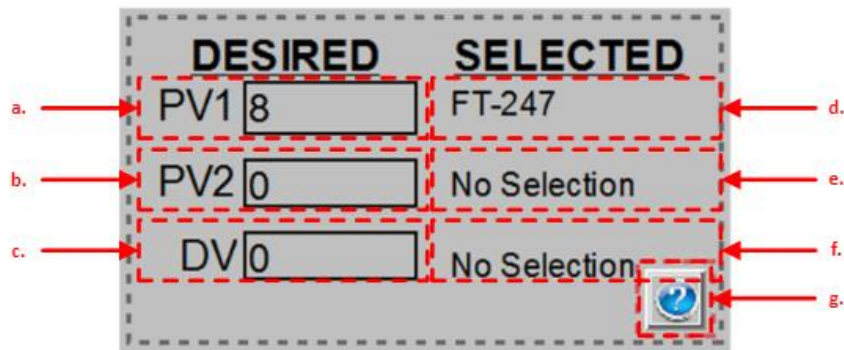


Figure 22: Variable selection section of the controller settings popup menu.

The following descriptions refer to Figure 22:

- PV1 selection field** – This field is used to select process variable one, selectable by entering the number 1-27; the corresponding assets associated with these numbers can be found in the PV, DV selection information popup (g.).
- PV2 selection field** – This field is used to select process variable two, selectable by entering the number 1-27; the corresponding assets associated with these numbers can be found in the PV, DV selection information popup (g.).

- c. **DV selection field** – This field is used to select the disturbance variable, selectable by entering the number 1-27; the corresponding assets associated with these numbers can be found in the PV, DV selection information popup (g.).
- d. **PV1 display** – Displays the current PV1 selected.
- e. **PV2 display** – Displays the current PV2 selected.
- f. **DV display** – Displays the current DV selected.
- g. **PV, DV Selection Information pushbutton** – Pulls up the PV, DV Selection Information popup.

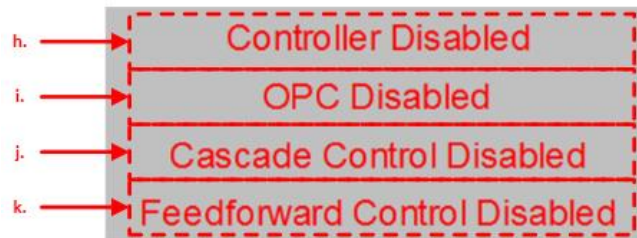


Figure 23: Controller configuration display section of the controller settings popup menu.

The following descriptions refer to Figure 23:

- h. **Controller status display** – Indicates the controller's state where 'Controller Disabled' indicates that the manipulated variable is in manual mode, and 'Controller Active' indicates the controller is actuating the manipulated variable.
- i. **OPC status display** – Indicates the state of OPC control where 'OPC Active' indicates the outer-loop controller is coming from a third-party program via OPC.
- j. **Cascade status display** – Indicates the status of cascade control where 'Cascade Control Active' indicates a cascaded configuration of the controller. *NOTE: Enabling cascade control while OPC is active feeds the output of the OPC outer-loop controller into the input of the EPKS-based inner-loop controller.*
- k. **Feedforward status display** – Indicates the status of feedforward control where 'Feedforward Control Active' indicates a feedforward configuration of the controller. *NOTE: Enabling feedforward control while OPC is active has NO effect.*

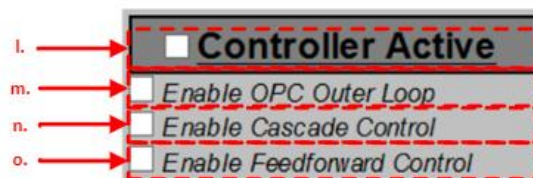


Figure 24: Controller configuration section of the controller settings popup menu.

The following descriptions refer to Figure 24:

- l. **Activate controller checkbox** – Used to set the controller to automatic mode.
- m. **Activate OPC checkbox** – Used to enable OPC outer-loop control.

- n. **Activate Cascade control checkbox** – Used to enable cascade control.
- o. **Feedforward control checkbox** – Used to enable feedforward control.

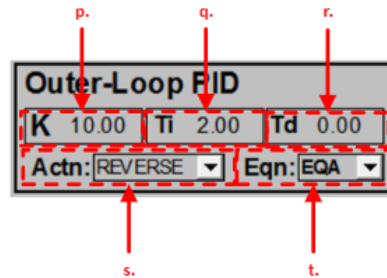


Figure 25: Outer-loop PID configuration section of the controller settings popup menu.

The following descriptions refer to Figure 25:

NOTE: Although the following sections refer to the outer-loop PID configuration fields, the same descriptions apply to the inner-loop PID configuration.

- p. **Controller Gain (Kc)** – This field sets the controller gain (Kc).
- q. **Controller integrator time** – This field sets the controller integrator time (Ti) in minutes.
- r. **Controller derivative time** – This field sets the controller derivative time (Td) in minutes.
- s. **Controller Action** – This field is used to set the sign of error, where in reverse mode ($Err = SP - PV$) and in direct mode ($Err = PV - SP$).
- t. **PID Equation display** – This is a read-only field; see section 3.6 Altering PID Equation for information on how to alter the PID equation.

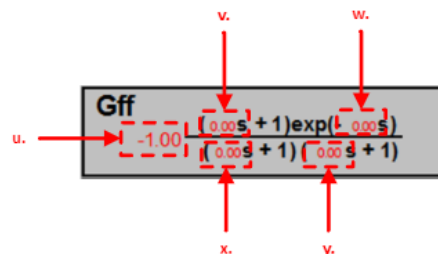


Figure 26: Feedforward controller equation configuration section of the controller settings popup menu.

The following descriptions refer to Figure 26:

- u. **Gff gain** – This field is used to set the gain of the feedforward controller.
- v. **Gff lead time constant** – This field is used to set the time constant of the lead system; set this parameter to zero to remove the lead system.
- w. **Gff delay** – This field is used to set the time delay of the feedforward controller; set this parameter to zero for a delay of zero seconds.
- x. **Gff lag time constant one** – This field is used to set the time constant of the first lag system; set this parameter to zero to remove the first lag system.
- y. **Gff lag time constant two** – This field is used to set the time constant of the second lag system; set this parameter to zero to remove the second lag system.

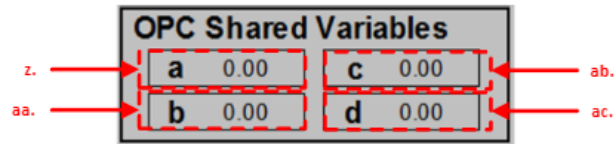


Figure 27: OPC shared variables section of the controller settings popup menu.

The OPC-shared variables are used as parameters that can be written to by the station HMI and read by third-party OPC-compliant programs. The intent is that these parameters are used to manipulate controller tuning parameters in programs operating outside of EPKS. Each of the four fields in Figure 27 (*z*, *aa*, *ab*, *ac*) correspond to a different OPC shared variable housed in the controller's CM.

3.6 Altering PID Equation

Changing the PID equation EPKS utilizes in each of the MUPPs controllers is possible. A description of the different PID equations can be found in Appendix A. However, this section will outline the process of changing the PID equation. The process will be demonstrated by changing the PID equation for the controller responsible for manipulating the cyclone underflow pump (CUFP-361). We only show the operation of altering a single controller; however, the process is identical for all other controlled parameters.

1. Using the search for Point-ID function, open the CM of the controller containing the PID, which we desire to change the equation for (Figure 28). *NOTE: For a complete list Point-IDs for all of the controllers of the MUPP, see Table 2 and Table 3.*
2. Deactivate the CMs 'EX ST' state (Figure 29).
3. Open the PID block parameters by double-clicking the function block.
4. Change the PID equation from the 'Algorithm' tab of the parameters menu (Figure 29).

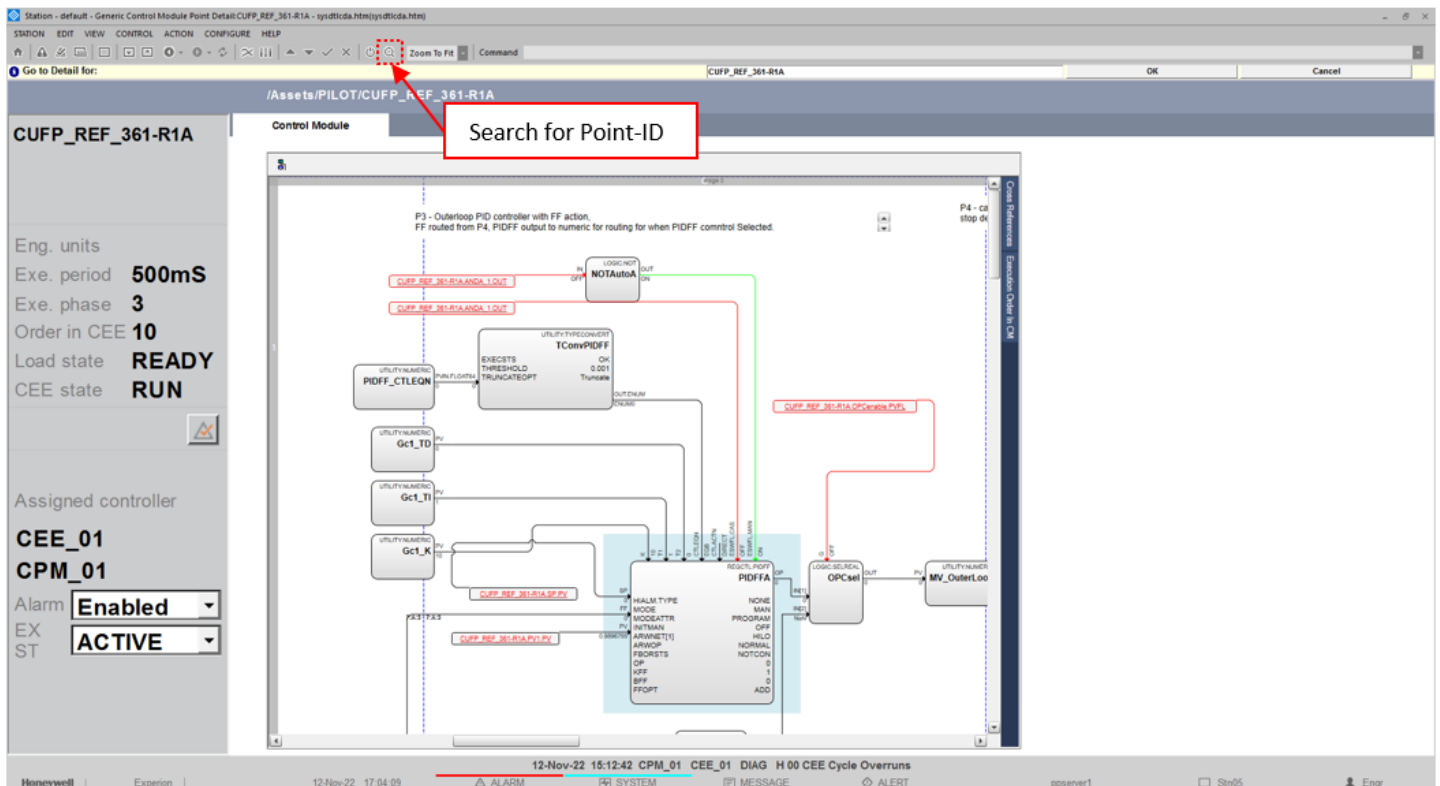


Figure 28: Opening a CM using the search for Point-ID function from the Experion Station toolbar.

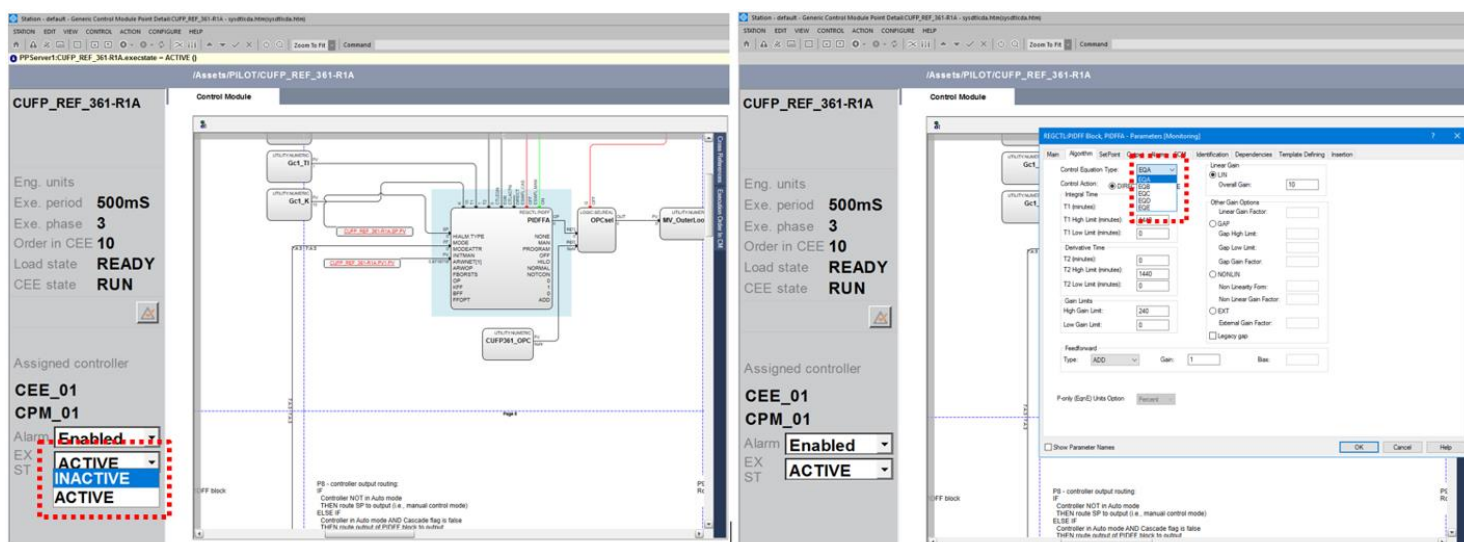


Figure 29: (LEFT) Setting the CMs 'EX ST' state to inactive. (RIGHT) Changing the PID equation of the controller.

Table 2: Point-IDs of the CMs containing the pump-associated controllers.

Asset	Point-ID
Ball Mill Pump	BMP_REF_241-R1A
Feed Pump	FP_REF_141-R1A
Cyclone Recycle Pump	CRP_REF_341-R1A
Cyclone Underflow Pump	CUPP_REF_361-R1A
Lamella Underflow Pump	LUP_REF_421-R1A
Feed Disturbance Pump	FDP_REF_521-R1A
Needle Tank Underflow Pump	NTP_REF_561-R1A
Product Pump	PP_REF_681-R1A

Table 3: Point-IDs of the CMs containing the flow control valve-associated controllers.

Asset	Point-ID
FCV-541	FCV_541-R1A
FCV-570	FCV_570-R1A
FCV-642	FCV_642-R1A
FCV-662	FCV_662-R1A
FCV-688	FCV_688-R1A
FCV-690	FCV_690-R1A

3.7 Interlock information

For the safety and longevity of the MUPP, several interlocks are programmed into the plant; these interlocks cannot be turned off and can only be removed by changing the configuration of the CMs loaded onto the controller from the EPKS control building software. The interlocks will take effect regardless of whether the plant is controlled from Experion Station or using third-party software over OPC. A list of all the interlocks that exist on the MUPP can be seen in Table 4. Further, a complete list of interlocks and system limits can be pulled up from the MUPP home screen in the Experion station.

Table 4: A complete account of the MUPP interlocks.

Asset	CM	Interlock ID	Interlock Conditions		On-Delay (s)
AG-101	AG_101	AG-101-INT-01		LT_102.LT_102.PV < 5%	0
AG-321	AG_321	AG-321-INT-01		LT_322.LT_322.PV < 5%	0
AG-661	AG_661	AG-661-INT-01		LT_667.LT_667.PV < 5%	0
AG-121	AG_OFF_121	AG-121-INT-01		LT_122.LT_122.PV < 5%	0
AG-221	AG_OFF_221	AG-221-INT-01		LT_222.LT_222.PV < 5%	0
		AG-221-INT-02		LT_222.LT_222.PV > 95%	0
BMP-241	BMP_OFF_241	BMP-241-INT-01		LT_222.LT_222.PV < 5%	0
		BMP-241-INT-02		LT_322.LT_322.PV > 100%	300
		BMP-241-INT-03	AND	BMP_OFF_241.BMP_RUN_241.PVFL == TRUE FT_247.DACA.PV < 5%	5
CRP-341	CRP_OFF_341	CRP-341-INT-01	AND	3WV_368.P2_368.PVFL == TRUE	300
				3WV_368.P1_368.PVFL == FALSE	
				LT_102.DACA.PV > 100	
		CRP-341-INT-02	AND	3WV_368.P2_368.PVFL == FALSE	300
				3WV_368.P1_368.PVFL == TRUE	
				LT_122.DACA.PV > 100	
CUFP-361	CUFP_OFF_361	CUFP-341-INT-01	AND	FT_363.DACA.PV < 5	5
				CUFP_OFF_361.CUFP_RUN_361.PVFL == TRUE	
		CUFP-341-INT-02		LT_222.DACA.PV > 100	300
		CUFP-351-INT-03		LT_322.DACA.PV < 5	0
DP-611	DP_OFF_611	DP-611-INT-01	AND	SV_614.FLAGA.PVFL == FALSE	0
				SV_615.FLAGA.PVFL == FALSE	
				SV_616.FLAGA.PVFL == FALSE	
		DP-611-INT-02	AND	FT_613.DACA.PV < 3 DP_OFF_611.DP_RUN_611.PVFL == TRUE	5

Asset	CM	Interlock ID	Interlock Conditions			On-Delay (s)
FDP-521	FDP_OFF_521	FDP-521-INT-01	LT_501.DACA.PV > 100%			300
		FDP-521-INT-02	AND	OR	FT_523.DACA.PV < 5%	12
					FDP_OFF_521.FDP_RUN_521 == TRUE	
					FDP_REF_521-R1A.FDP_521.PV > 0	
				AND	FDP_REF_521-R1A.AUTO.PVFL == TRUE	
					FDP_REF_521-R1A.AUTO.PVFL == FALSE	
				AND	FDP_REF_521.FDP_521.PV > 0	
					FDP_REF_521.PD_12_6_SWM.PVFL == FALSE	
					FDP_REF_521.PD_12_6_SWM.PVFL == TRUE	
		FDP-521-INT-03	LT_542.DACA.PV < 5%			0
FP-141	FP_OFF_141	FP-141-INT-01	AND			0
		FP-141-INT-02	AND			0
		FP-141-INT-03	LT_222.DACA.PV > 100%			300
		FP-141-INT-04	AND			5
LUP-421	LUP_OFF_421	LUP-421-INT-01	LT_422.LA_LL.PV == FALSE			0
NTP-561	NTP_OFF_561	NTP-561-INT-01	LT_501.DACA.PV < 5%			0
		NTP-561-INT-02	AND			5
PP-681	PP_OFF_681	PP-681-INT-01	AND			5
		PP-681-INT-02	LT_667.DACA.PV < 5%			0
SV-104	SV-104	SV-104-INT-01	LT_102.LT_102.PV > 102%			0
SV-105	SV-105	SV-105-INT-01	LT_102.LT_102.PV > 102%			0
SV-124	SV-124	SV-124-INT-01	LT_122.LT_122.PV > 102%			0
SV-125	SV-125	SV-125-INT-01	LT_122.LT_122.PV > 102%			0
SV-614	SV-614	SV-614-INT-01	LT_667.LT_667.PV > 102%			0
SV-615	SV-615	SV-615-INT-01	LT_667.LT_667.PV > 102%			0
SV-616	SV-616	SV-616-INT-01	LT_667.LT_667.PV > 102%			0
SV-691	SV-691	SV-691-INT-01	LT_667.LT_667.PV > 102%			0
SV-692	SV-692	SV-692-INT-01	LT_667.LT_667.PV > 102%			0
SV-693	SV-693	SV-693-INT-01	LT_667.LT_667.PV > 102%			0

3.8 Trend Pages

The trend pages monitor groups of parameters from the Experion HMI. There are two methodologies for pulling up trend pages:

1. By selecting 'request trend' from the station toolbar before entering the number of the trend-group, you wish to view (from 1-3000) (Figure 30).
2. By opening the view dropdown menu and selecting trends (Figure 31), the complete list of trends groups will be displayed (Figure 32).

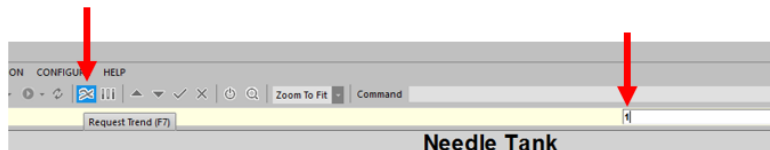


Figure 30: Opening a trend group using the request trend function.

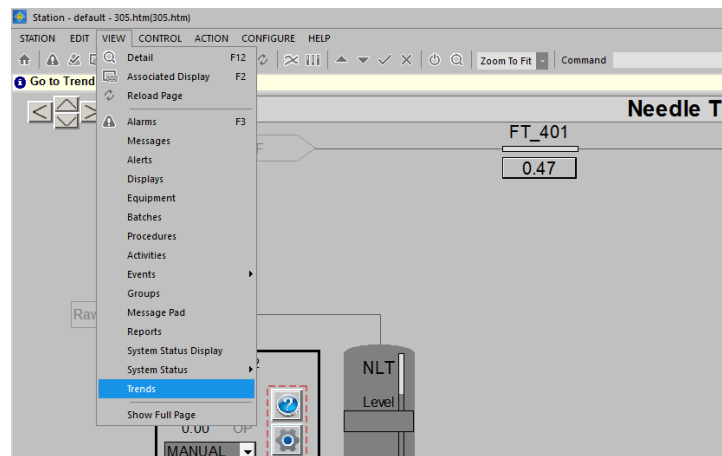


Figure 31: Pulling up the trend group selection screen using the 'View' dropdown menu.

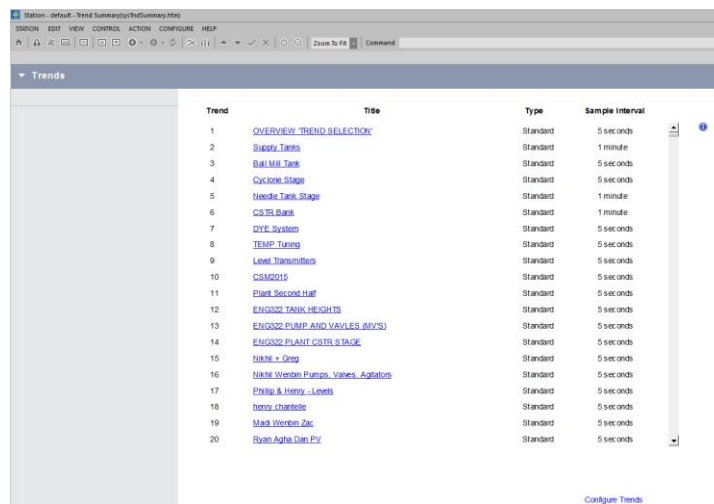


Figure 32: The station trend-group selection screen.

With a trend group open, data can be exported to Excel by right clicking the trend and selecting copy (Figure 33). Once copied, it can be pasted directly into Excel (Figure 34).

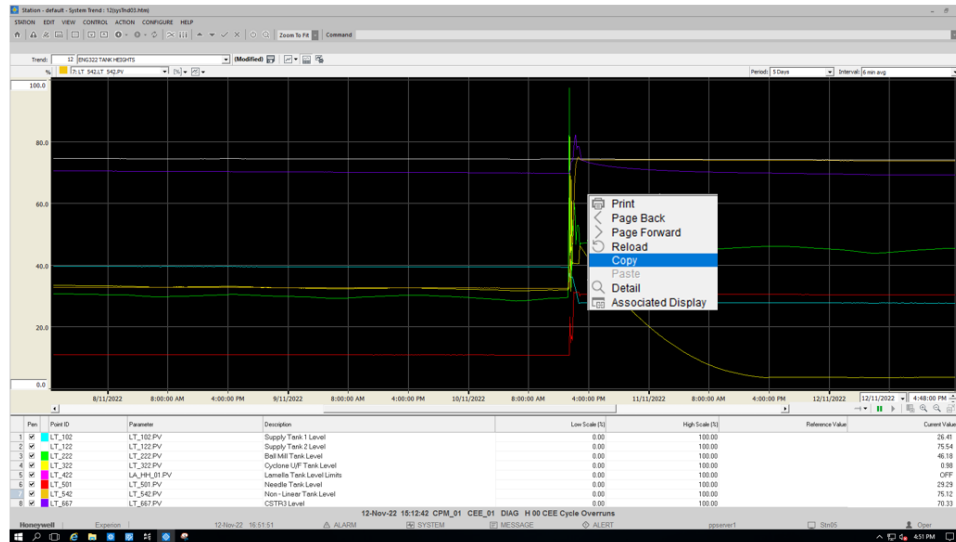


Figure 33: Copying data from a station trend group page.

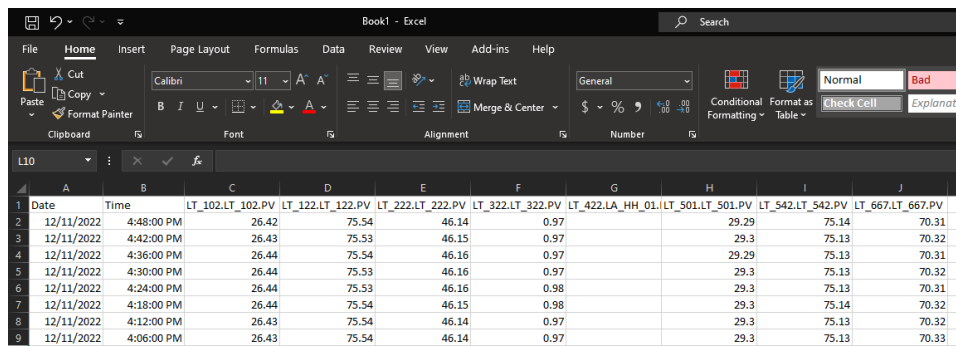


Figure 34: Station trend group data copied from a station trend group page and pasted into Microsoft Excel.

4.0 Additional Reading

Consult Table 5 for recommendations for further reading.

Table 5: Recommended further reading and their relevance in the context of this document.

<i>Document:</i>	<i>Relevance:</i>
PP-1001: OPC Tag Names & Communication Rev.1	Useful for uncovering Point-ID and Point Parameter information for the variables available on the MUPP OPC server.
PP-5007: 'OPCControlTemplateR0': Use and Function Rev. 0	Outlines how to implement parameter control from MATLAB using the 'OPCControlTemplateR0' Program.

Appendix A: PID FB Algorithms

The PID block used for function block programming in the C300 has four different equations that can be selected for calculating the PID algorithm (Equation A to Equation D).

By default Equation A is selected however the option to change the equation is available in station.

Equation A

This equation is formally known as the Parallel incremental algorithm and differs from the classical PID algorithm of Ogunnaike & Ray 1994: Process Dynamics, Modeling, and Control, in that the error term is reversed and the derivative term is subject to a first order filter with gain of unity and time constant equal to $(\alpha * T2)$.

$$CV = K * L^{-1} \left[\left(1 + \frac{1}{T1_s} + \frac{T2_s}{1 + \alpha * T2_s} \right) * (PVP_s - SPP_s) \right]$$

Equation B:

Equation B differs in that the derivative term acts on the changes in PV while the proportional and integral terms continue to act on the error (PV – SP). This equation is beneficial when quick SP changes are implemented because with the derivative term acting on the changes in PV, the effect of derivative spikes are minimized.

$$CV = K * L^{-1} \left[\left(1 + \frac{1}{T1_s} + \frac{T2_s}{1 + \alpha * T2_s} \right) * PVP_s - \left(1 + \frac{1}{T1_s} \right) * SPP_s \right]$$

Equation C:

Equation C produces the smoothest response to SP changes, this is because unlike equations A and B, both the proportional and derivative terms act on changes in PV. The integral term remains acting on the error (PV – SP) and ensures minimal steady state offset.

$$CV = K * L^{-1} \left[\left(1 + \frac{1}{T1_s} + \frac{T2_s}{1 + \alpha * T2_s} \right) * PVP_s - \left(\frac{1}{T1_s} \right) * SPP_s \right]$$

Equation D:

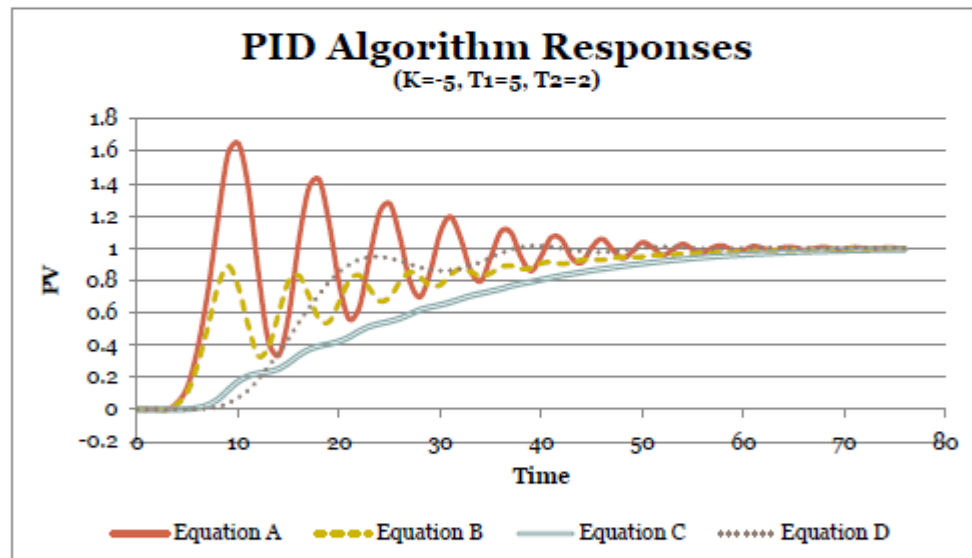
Equation D differs from all other equations in that it provides only integral control.

$$CV = L^{-1} \left[\frac{1}{T1_s} * (PVP_s - SPP_s) \right]$$

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Acronyms used in PID equations:	
CV	Output of PID in percent
K	Gain (proportional term)
L^{-1}	Inverse of the LaPlace transform
PV	Process input value in engineering units
PVP	PV in percent
α	1/16 fixed filter constant.
s	LaPlace operator
SPP	Set Point in percent
T ₁	Integral time constant in minutes
T ₂	Derivative time constant in minutes

The PID algorithm responses presented below illustrate the effect of each controller, this simulation was presented against an underdamped second order system.



The relationship between the parameters in Equation A and the classical algorithm are:

$$Kc = -K \quad \tau_I = T_1 \quad \&$$

$$T_2 = \frac{\tau_D}{1 - \alpha \tau_D} \text{ or } \frac{T_2}{1 + \alpha T_2}$$

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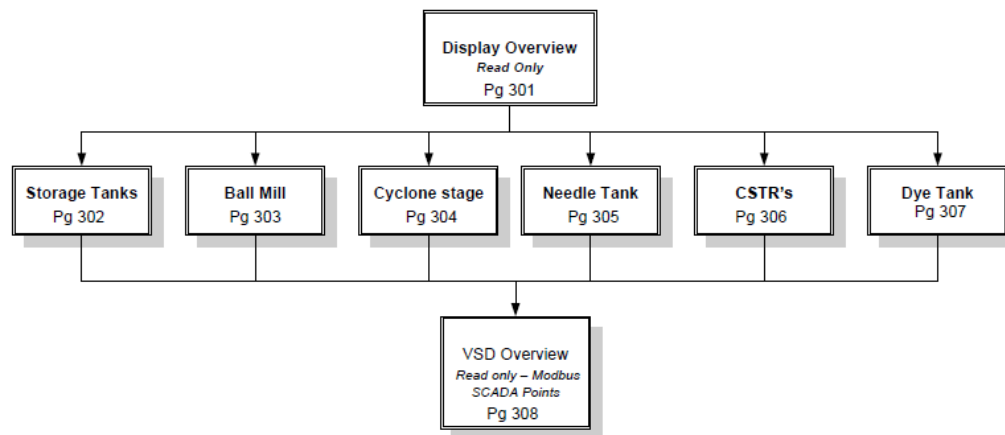
Appendix B: Display Hierarchy and Control Levels

The displays are arranged in a hierarchical fashion that provides the user with a top level overview display that monitors all vitals of the entire plant.

Beneath this display sits six detailed control displays. These displays focus on individual unit operations and provide the user with manual and automatic control.

The lowest level is another monitoring display. This display monitors the VSD's over the Modbus link. This display is unique from all others in that no data is read through the C300, but only over the SCADA Modbus link. Thus if the C300 drops out then these vitals can still be received.

The display hierarchy is illustrated below.



Each display has a unique page number. This page number if searched in Station will bring up the desired display, alternatively each page is hyperlinked to related displays.

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